

Labels warning about alcohol-attributable cancer risks should be mandated urgently



In January, 2025, the US Surgeon General released an Advisory on alcohol consumption and cancer risk.¹ He briefly summarised the evidence on alcohol and cancer, including underlying biological mechanisms, and concluded that alcohol use is a leading preventable cause of cancer in the USA and globally, causing around 100 000 and 750 000 cancer cases annually, respectively.² The carcinogenicity of alcohol is not a new concept. More than three decades ago, the International Agency for Research on Cancer (IARC) and the Continuous Update Project of the World Cancer Research Fund/American Institute for Cancer Research concluded that there was sufficient evidence that alcohol causes certain cancers;³ the current list of alcohol-attributable cancer sites published by IARC⁴ includes cancers of the oral cavity, oropharynx, hypopharynx, oesophagus (squamous cell carcinoma), colon, rectum, liver, and intra-hepatic bile duct, larynx, and female breast.

The novel aspect of the Surgeon General Advisory was its conclusion and call for action. The advisory recommended that health warning labels should be updated on alcoholic beverages as the next important step for the prevention of alcohol-attributable cancer.¹ This guidance comes after Ireland mandated the placement of cancer warnings on all alcoholic beverage labels to start in May, 2026. Worldwide, Ireland is only the second country after South Korea to legislate the inclusion of cancer warning messages on alcohol labels. In South Korea, however, the manufacturers can opt to apply alternative labels that do not mention cancer on their products. The 2022–30 WHO Global Alcohol Action Plan called on the Member States to “ensure appropriate consumer protection measures through development and implementation of labelling requirements for alcoholic beverages that display essential information for health protection”, and the European Union has commissioned reports sharing the same conclusion: cancer warning labels, if properly designed, will be an effective way to inform customers about the carcinogenic properties of alcohol.⁵

Information for consumers is key, as to date the general population is not sufficiently informed about the cancer risks of alcohol, despite all the available

evidence. Studies have consistently highlighted limited awareness among the general population about the association between alcohol and cancer globally⁶ with less than half of people in most of the studied countries aware of the connection. Awareness tends to be lower if respondents are asked to list the most common risk factors for cancer, as opposed to selecting them from a pre-determined list, highlighting the lack of salience regarding the association in peoples’ minds. Another consistent finding is that there is lower awareness for certain specific types of cancer linked to alcohol—while most people do associate alcohol with liver cancer, the association between alcohol and breast cancer, for example, is much less recognised; in some countries, less than 10% of people surveyed were aware of the association.^{6,7}

Disseminating information regarding the cancer risks of alcohol is especially imperative since this risk has no lower threshold.⁸ For example, a 2024 systematic review and meta-analysis of all studies on alcohol use and female breast cancer showed that for consumption of less than one standard drink (10 g pure alcohol per day), the relative risk estimate for breast cancer was significantly elevated at 1.04 (95% CI 1.01–1.07) compared with the counterfactual scenario of not drinking.⁹ Such information is important, together with absolute risk information, since many women consume alcohol at lower levels than men.¹⁰ Consequently, relatively large proportions of alcohol-attributable cancer (14%) are caused by drinking fewer than two drinks a day globally; in absolute numbers, this amounts to more than 100 000 cases a year.²

In the past two decades, alcohol producers have strongly opposed the introduction of mandatory cancer warning labels on alcohol beverage containers in various jurisdictions.¹¹ Such opposition has even resulted in the termination of research examining the use of cancer warning labels.¹² Similar interference based on similar arguments are expected in response to public health efforts motivated by the US Surgeon General advisory, therefore public health advocates must be prepared to provide the relevant evidence to support proposed mandates.¹³

Published Online
February 13, 2025
[https://doi.org/10.1016/S2468-2667\(25\)00040-4](https://doi.org/10.1016/S2468-2667(25)00040-4)

For more on Ireland’s mandate see <https://www.irishstatutebook.ie/eli/2023/si/249/made/en/print>

For more on the WHO Global Alcohol Action Plan see <https://www.who.int/publications/iitem/9789240090101>

In conclusion, alcohol has been shown to be a leading cause of cancer with no lower threshold, causing a substantial number of cases even at low levels of alcohol consumption. Given that knowledge about the association between alcohol and cancer has been consistently low in the general population, warning labels seem to be a suitable way to inform consumers, and allow them to make informed decisions about the cancer risks associated with their alcohol consumption.

CF-B, MN, and GG are employees of WHO. DK and JR are WHO consultants. The authors alone are responsible for the views expressed here and these do not necessarily represent the decisions or the stated policy of WHO.

Copyright © 2025 The Author(s). Published by Elsevier Ltd. This is an Open Access article under the CC BY-NC-ND 4.0 license.

Carina Ferreira-Borges, Daša Kokole, Gauden Galea, Maria Neufeld, *Jürgen Rehm
jtrehm@gmail.com

WHO Regional Office for Europe, Copenhagen, Denmark (CF-B, DK, GG, MN); Department of Health Promotion, CAPHRI Care and Public Health Research Institute, Maastricht University, Maastricht, Netherlands (DK); Institute for Mental Health Policy Research, Campbell Family Mental Health Research Institute, and PAHO/WHO Collaborating Centre at CAMH, Centre for Addiction and Mental Health Toronto, ON M5T 1R8, Canada (JR); Institute of Health Policy, Management and Evaluation, Dalla Lana School of Public Health, University of Toronto, Toronto, ON, Canada (JR); Institute of Medical Science, Temerty Faculty of Medicine, Toronto, ON, Canada, M5S 3H2 (JR); Department of Psychiatry and Psychotherapy, Centre of Interdisciplinary Addiction Research, University Medical Center Hamburg-Eppendorf, Hamburg, Germany (JR); Program on Substance Abuse & WHO European Region Collaborating Centre, Public Health Institute of Catalonia, Barcelona, Spain (JR)

- 1 US Department of Health and Human Services. Alcohol and cancer risk, 2025. The US Surgeon General's Advisory. <https://www.hhs.gov/surgeongeneral/reports-and-publications/alcohol-cancer/index.html> (accessed Feb 6, 2025).

- 2 Rumgay H, Shield K, Charvat H, et al. Global burden of cancer in 2020 attributable to alcohol consumption: a population-based study. *Lancet Oncol* 2021; **22**: 1071–80.
- 3 International Agency for Research on Cancer. IARC monographs on the evaluation of the carcinogenic risks to humans. Alcohol drinking. <https://publications.iarc.fr/62> (accessed Feb 6, 2025).
- 4 International Agency for Research on Cancer. IARC monographs on the evaluation of the carcinogenic risks to humans. Personal habits and indoor combustions. <http://publications.iarc.fr/122> (accessed Feb 6, 2025).
- 5 European Commission. Health warnings on alcoholic beverages in EU Member States. November, 2024. <https://data.europa.eu/doi/10.2925/8649276> (accessed Jan 26, 2025).
- 6 Kokole D, Ferreira-Borges C, Galea G, Tran A, Rehm J, Neufeld M. Public awareness of the alcohol-cancer link in the EU and UK: a scoping review. *Eur J Public Health* 2023; **33**: 1128–47.
- 7 Neufeld M, Kokole D, Correia D, et al. How much do Europeans know about the link between alcohol use and cancer? Results from an online survey in 14 countries. *BMC Res Notes* 2024; **17**: 56.
- 8 Anderson BO, Berdzuli N, Ilbawi A, et al. Health and cancer risks associated with low levels of alcohol consumption. *Lancet Public Health* 2023; **8**: e6–7.
- 9 Sohi I, Rehm J, Saab M, et al. Alcoholic beverage consumption and female breast cancer risk: a systematic review and meta-analysis of prospective cohort studies. *Alcohol Clin Exp Res* 2024; **48**: 2222–41.
- 10 WHO. Global status report on alcohol and health and treatment of substance use disorders. <https://www.who.int/publications/i/item/9789240096745> (accessed Feb 6, 2025).
- 11 Stockwell T, Solomon R, O'Brien P, Vallance K, Hobin E. Cancer warning labels on alcohol containers: a consumer's right to know, a government's responsibility to inform, and an industry's power to thwart. *J Stud Alcohol Drugs* 2020; **81**: 284–92.
- 12 Babor TF. The arrogance of power: alcohol industry interference with warning label research. *J Stud Alcohol Drugs* 2020; **81**: 222–24.
- 13 WHO. Empowering public health advocates to navigate alcohol policy challenges: alcohol policy playbook. <https://www.who.int/europe/publications/i/item/WHO-EURO-2024-5624-45389-76520> (accessed Feb 6, 2025).